Phase III: “Software Desing and Modeling”

Explanation of UML Diagrams – Tourism Reservation software

1. System Architecture

The system is a web-based tourism reservation platform where tourists can browse events in Albania, reserve tickets, and receive confirmations.

The architecture follows a layered structure:

• Frontend/UI: User interface where tourists and admins interact with the system.

• Backend/Application Server (Node.js): Processes requests, applies business logic, validates users, handles reservations, and communicates with the database.

• Database (MySQL): Stores users, events, reservations, and ticket information.

• Email Service: Sends booking confirmations and ticket numbers.

• Security Module: Handles authentication, encryption, access control, and admin authorization.

All components work together so users can register/login, browse events, make reservations, and admins can create/manage events.

2. Component Diagram (UML)

Main components shown in your diagram:

• Tourism Reservation System: Central controller coordinating all modules.

• Restaurants: Stores restaurant reservation information for tourists.

• Tourist: Represents users who browse events and make bookings.

• Reservation: Handles booking creation, confirmation, and cancellations.

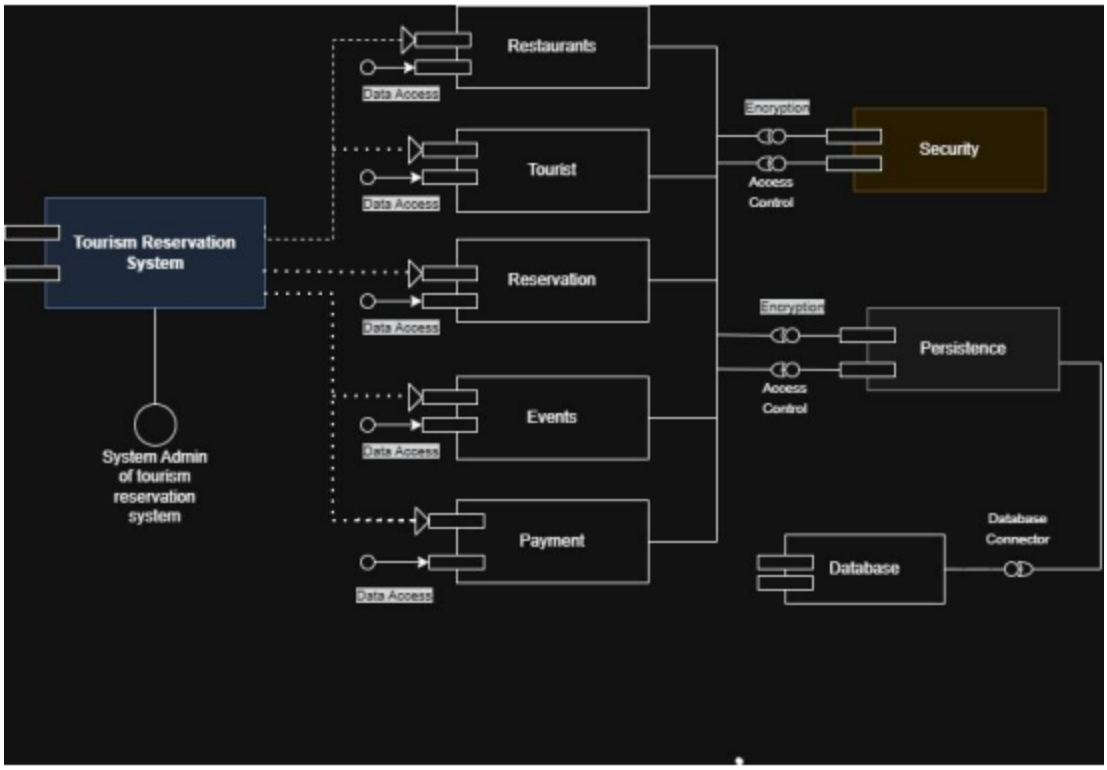
• Events: Manages event details such as title, date, location, and capacity.

• Payment: Responsible for payment processing (if enabled).

• Security: Provides encryption, access control, authentication, and authorization.

• Persistence: Manages saving and retrieving data.

• Database: Stores all permanent records.



Interactions:

• Tourist sends requests through the Tourism Reservation System.

• Reservation, Events, Restaurants, and Payment modules exchange data through Persistence.

• Security protects access to the system.

• Persistence connects directly to the Database.

3. Class Diagram

The class diagram contains three main classes:

User

Attributes: user_id, name, email, password, role

Purpose: Stores user accounts for tourists or admins.

Reservation

Attributes: reservation_id, ticket_number, user_id, event_id

Purpose: Represents each booking made by a user.

Event

Attributes: event_id, title, location, date, capacity

Purpose: Stores event information and available seats.

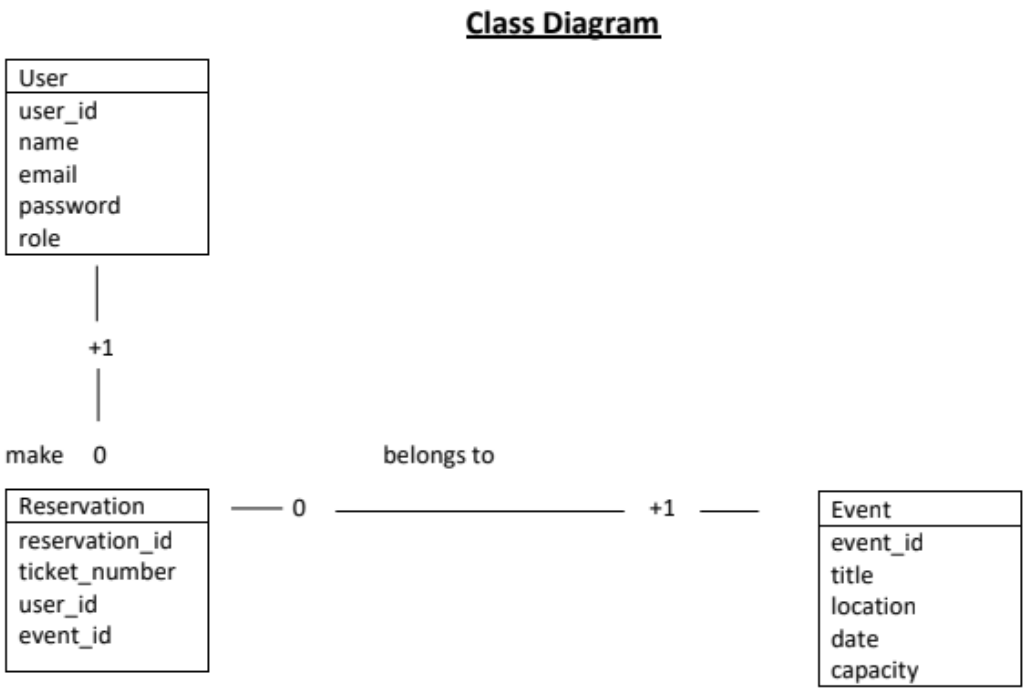
Relationships:

• One User can make many Reservations.

• Each Reservation belongs to one User.

• One Event can have many Reservations.

• Each Reservation is linked to one Event.



4. Sequence Diagrams

A. Ticket Reservation Sequence

1. Tourist browses events on the frontend.

2. Frontend requests event list from backend.

3. Backend retrieves events from database.

4. Tourist selects an event and quantity.

5. Backend checks remaining capacity.

6. If tickets are available:

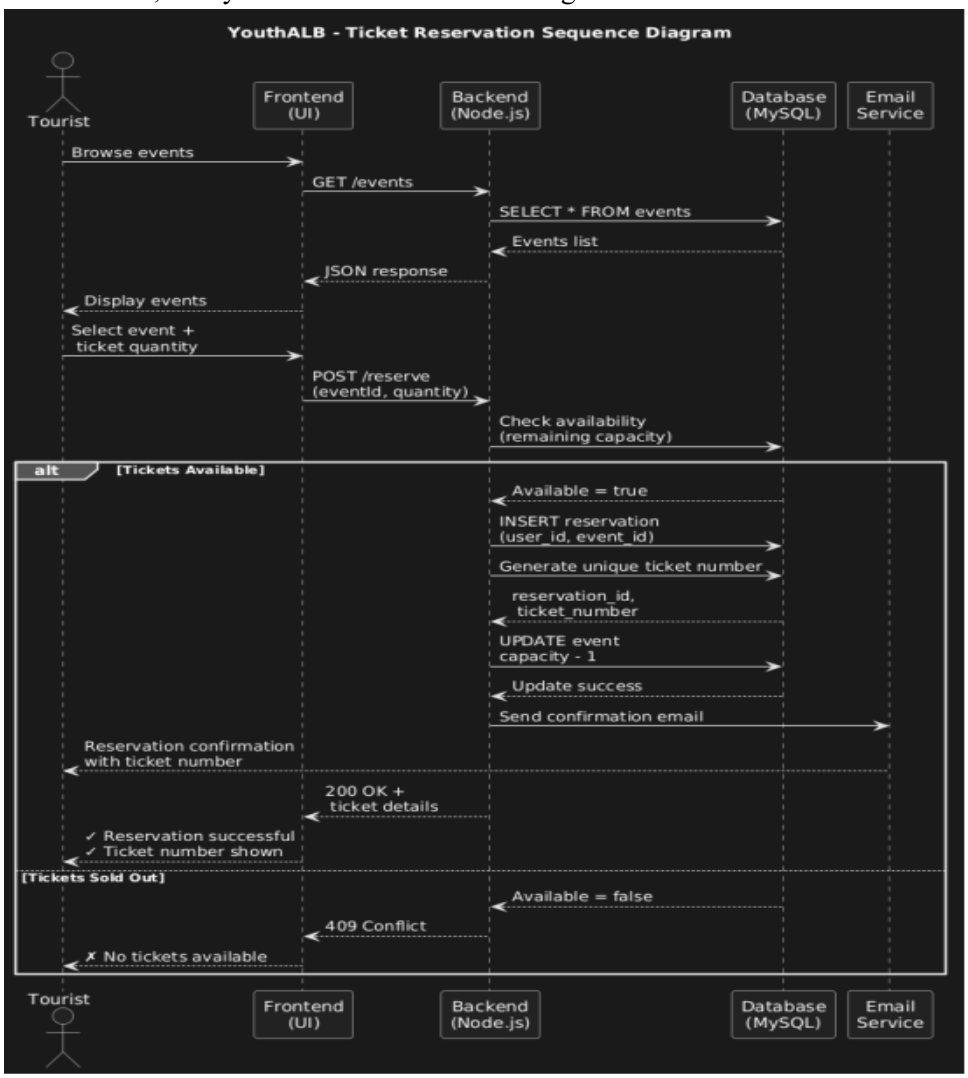
   – Reservation is inserted.

   – Unique ticket number is generated.

   – Event capacity is updated.

   – Confirmation email is sent.

7. If sold out, the system returns an error message



B. Admin Create Event Sequence

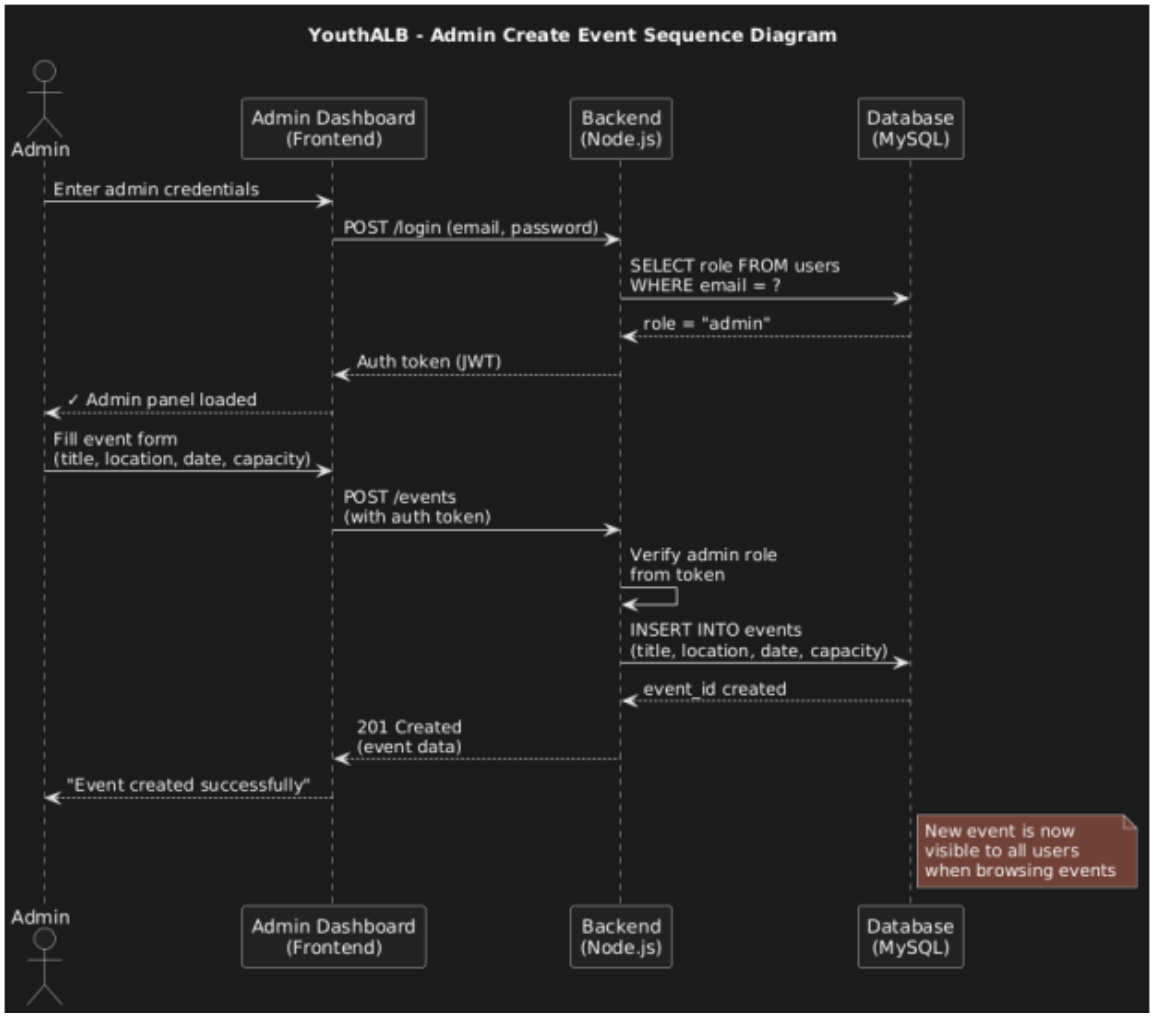
1. Admin logs in with credentials.

2. Backend verifies admin role.

3. Admin fills event form.

4. Backend stores new event in database.

5. Success response is returned and event becomes visible.



C. User Login Sequence

1. Tourist enters email and password.

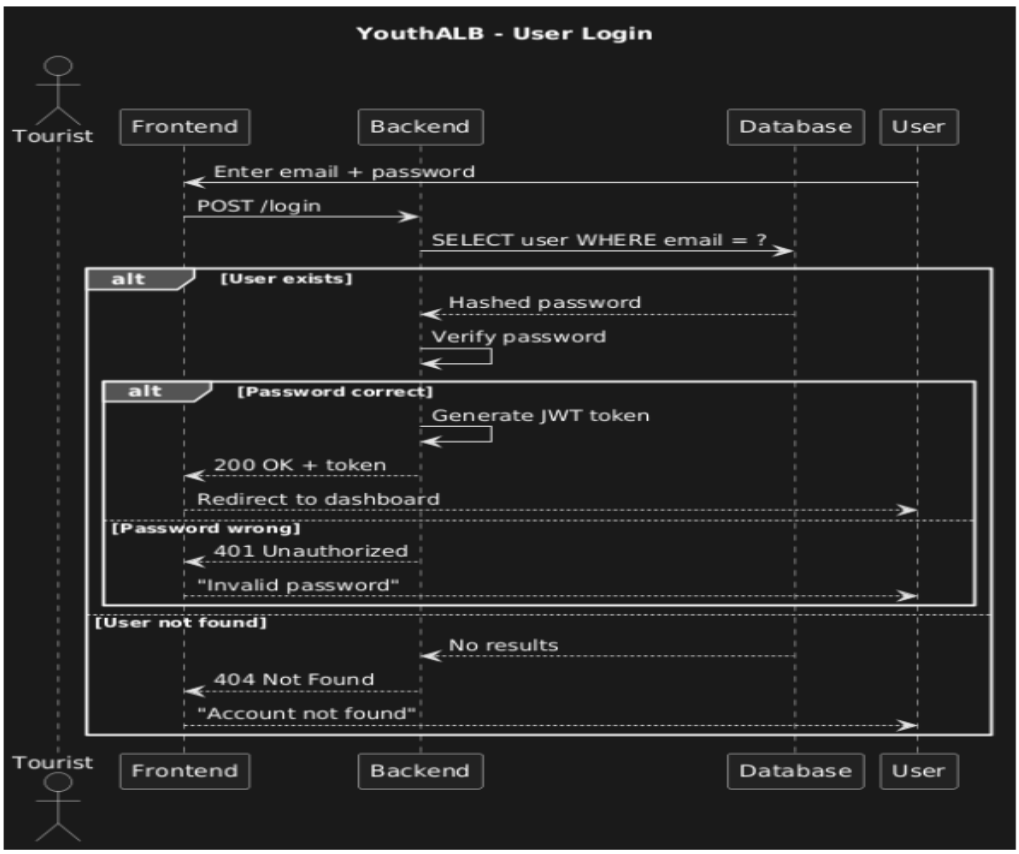
2. Backend searches for user.

3. Password hash is verified.

4. If correct, JWT token is generated and dashboard opens.

5. If wrong, unauthorized message is returned.

6. If user does not exist, account not found is shown.



5. Activity Diagrams

Main Booking Activity:

Start → Log in → Browse Events → Filter Events → Select Event → Check Ticket Availability

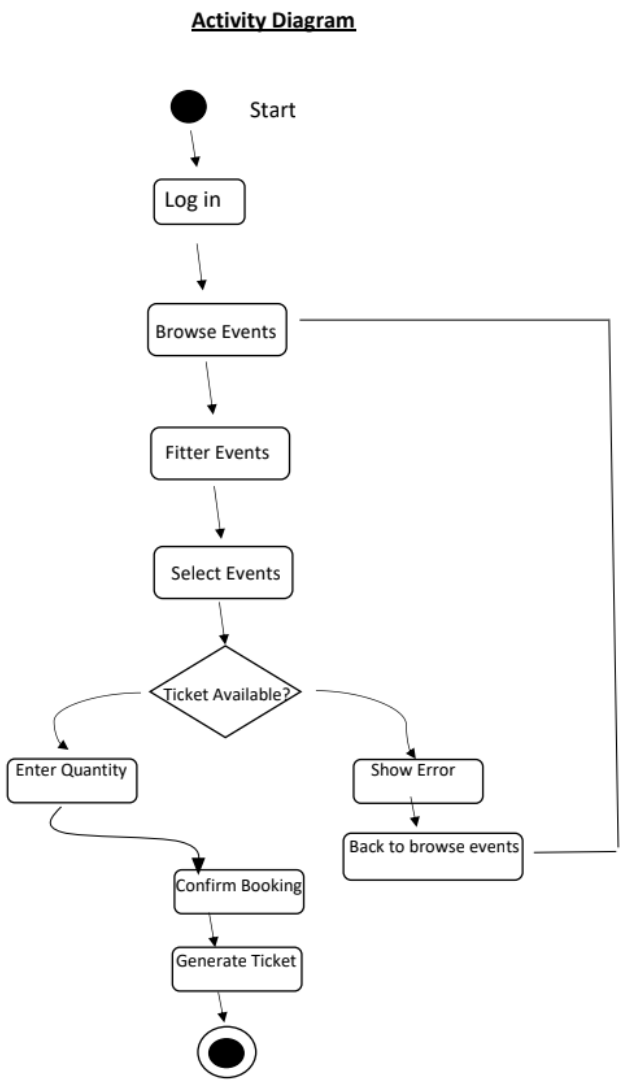
If tickets unavailable:

Show Error → Return to Browse Events

If tickets available:

Enter Quantity → Confirm Booking → Generate Ticket → End

This activity diagram shows the workflow a tourist follows to complete a reservation.



6. State Diagrams

A. Reservation State Diagram

Created → Pending → Confirmed → Completed

Alternative transitions:

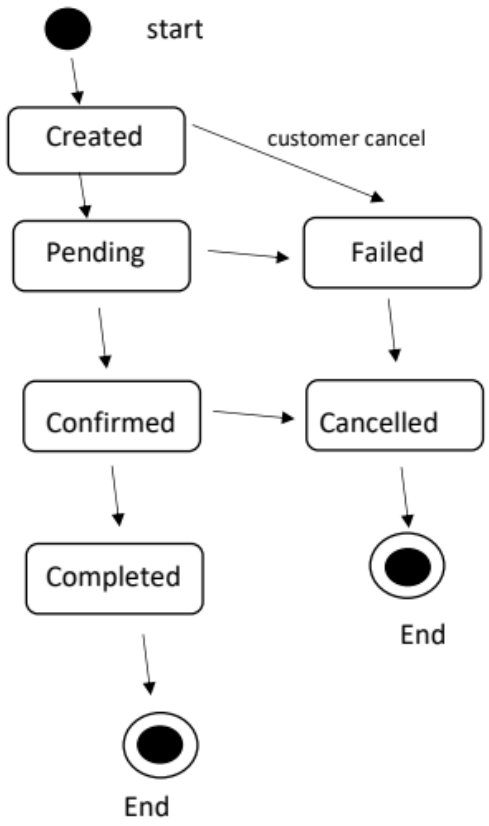
• Pending → Failed

• Created/Pending/Confirmed → Cancelled

This represents the life cycle of a reservation.

State Diagrams

1) Reservation State

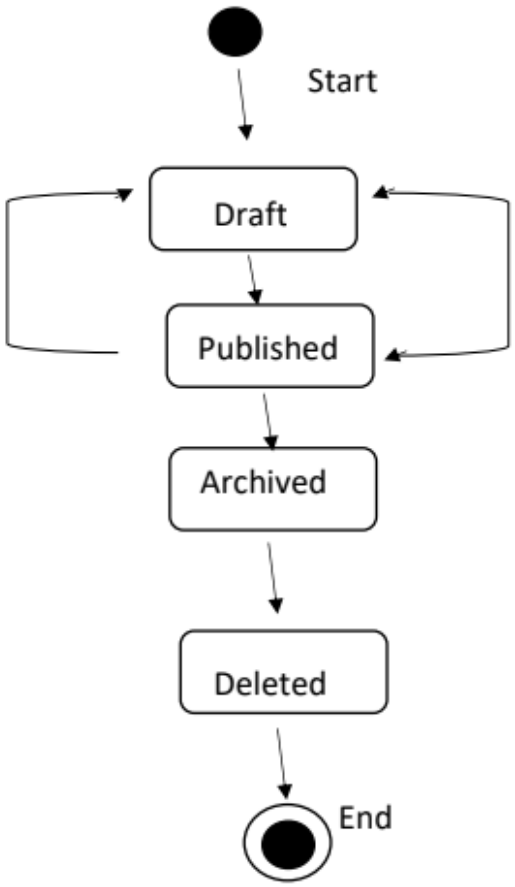


B. Event State Diagram

Draft → Published → Archived → Deleted

This shows how an event moves from creation to final removal.

2) Event Diagram



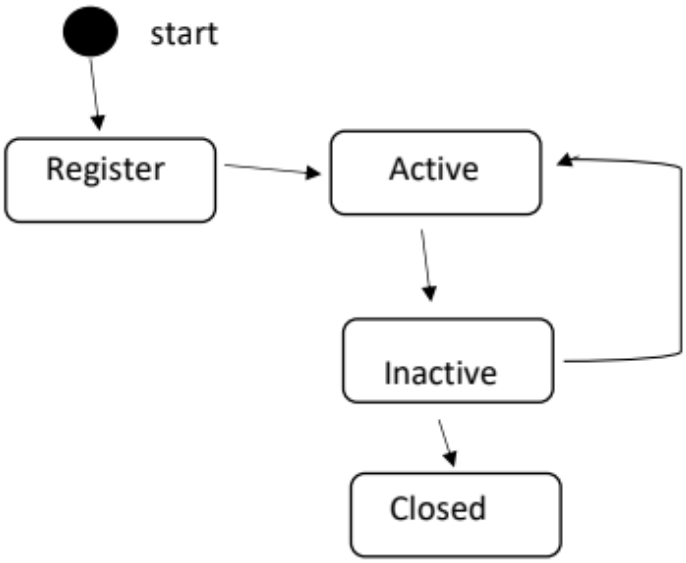
C. Activity Lifecycle Diagram

Register → Active → Inactive → Closed

Inactive may return to Active.

This represents the status of a user account or service activity over time.

2) Activity lifecycle



7. Use Case Diagram

The Use Case Diagram represents the functional requirements of the tourism reservation website by showing how users interact with the system.

Actors:

• User/Tourist: Main customer who uses the platform to browse and reserve tickets.
• Admin: System administrator who manages events and platform content.

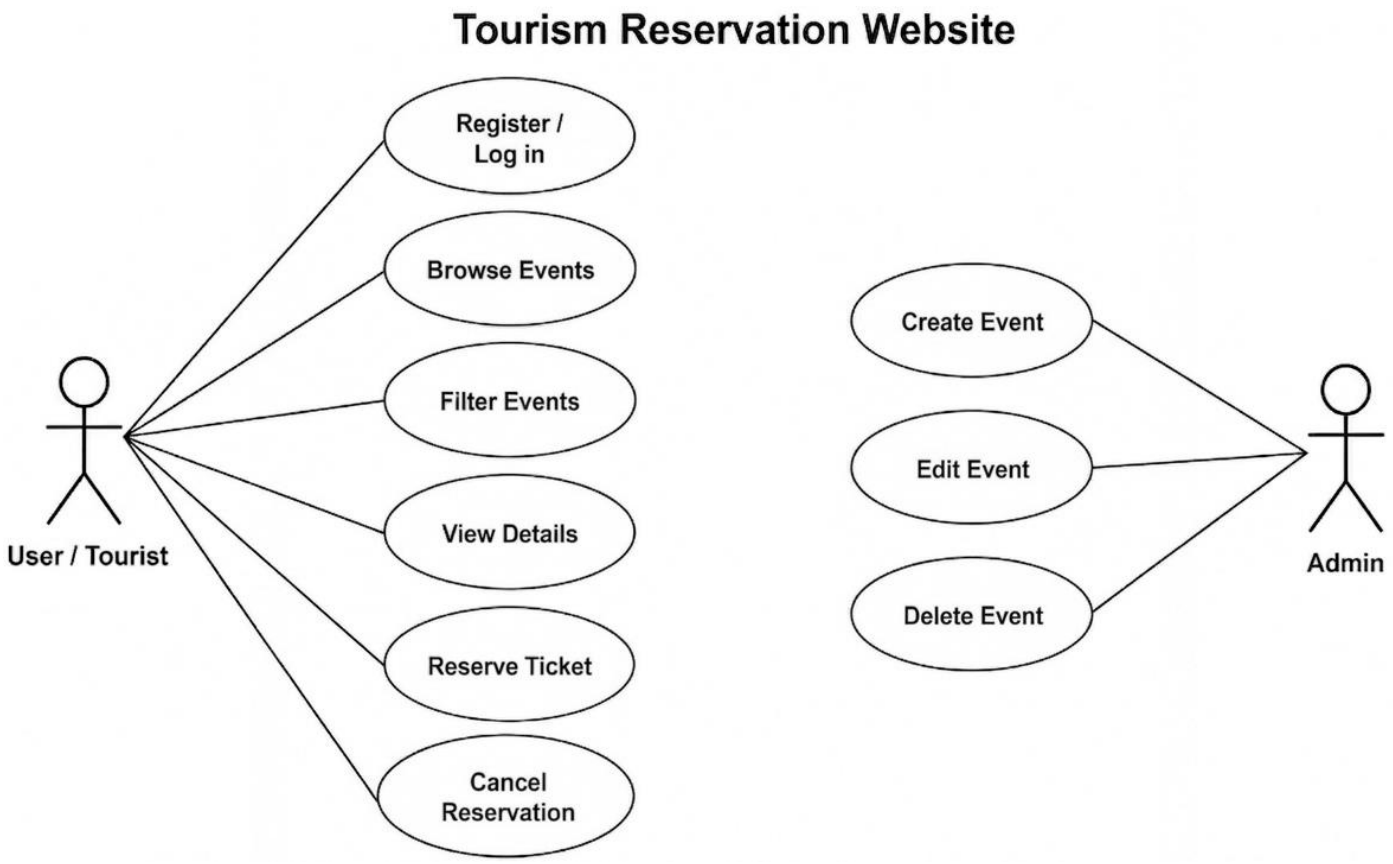
User Use Cases:

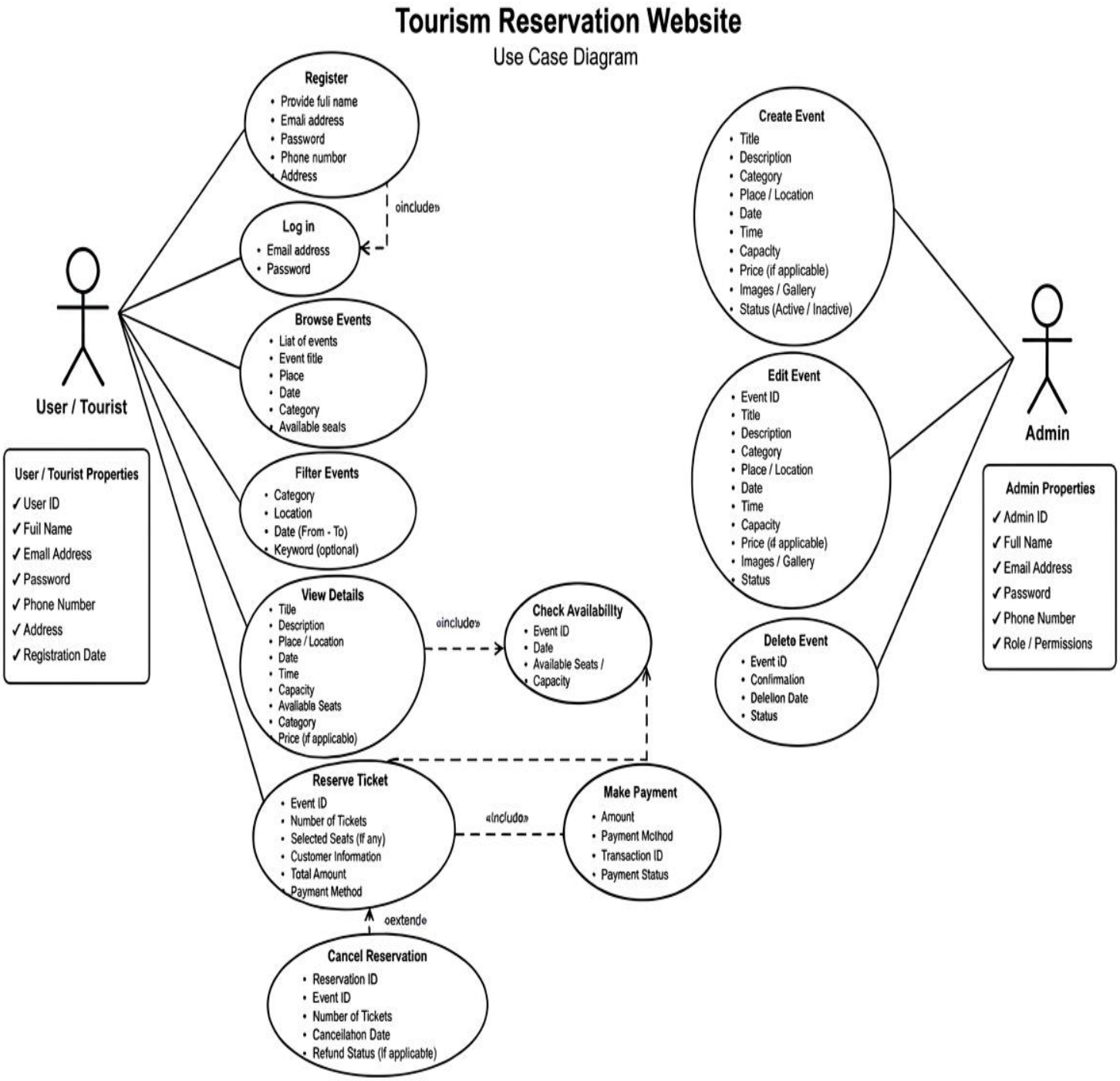
• Register / Log in: Create an account or access the system.
• Browse Events: View available events and tourism activities.
• Filter Events: Search events by category, location, or date.
• View Details: See full event information such as title, place, time, and capacity.
• Reserve Ticket: Book one or more tickets for an event.
• Cancel Reservation: Cancel an existing booking if allowed.

Admin Use Cases:

• Create Event: Add a new event to the platform.
• Edit Event: Modify event details such as date, title, or capacity.
• Delete Event: Remove an event from the system.

Explanation:




Tourism Reservation Website
Use Case Diagram
Register
• Provide full name
• Email address
• Password
• Phone number
• Address
«include»
Log in
• Email address
• Password
Browse Events
• List of events
• Event title
• Place
• Date
• Category
• Available seats
Filter Events
• Category
• Location
• Date (From - To)
• Keyword (optional)
View Details
• Title
• Description
• Place / Location
• Date
• Time
• Capacity
• Available Seats
• Category
• Price (if applicable)
«include»
Check Availability
• Event ID
• Date
• Available Seats /
• Capacity
Reserve Ticket
• Event ID
• Number of Tickets
• Selected Seats (If any)
• Customer Information
• Total Amount
• Payment Method
«include»
Make Payment
• Amount
• Payment Method
• Transaction ID
• Payment Status
«extend»
Cancel Reservation
• Reservation ID
• Event ID
• Number of Tickets
• Cancellation Date
• Refund Status (If applicable)
User / Tourist
User / Tourist Properties
✓ User ID
✓ Full Name
✓ Email Address
✓ Password
✓ Phone Number
✓ Address
✓ Registration Date
Create Event
• Title
• Description
• Category
• Place / Location
• Date
• Time
• Capacity
• Price (if applicable)
• Images / Gallery
• Status (Active / Inactive)
Edit Event
• Event ID
• Title
• Description
• Category
• Place / Location
• Date
• Time
• Capacity
• Price (if applicable)
• Images / Gallery
• Status
Delete Event
• Event ID
• Confirmation
• Deletion Date
• Status
Admin
Admin Properties
✓ Admin ID
✓ Full Name
✓ Email Address
✓ Password
✓ Phone Number
✓ Role / Permissions